

Jeffrey Wang

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RESEARCH INTERESTS

My research centers on rethinking the primitive building blocks of deep architectures and how new designs change what a model can learn and at what cost. My first-author work, PolyNeXt (ICML 2026), replaces standard nonlinearities with activation-free polynomial (Hadamard-product) modules that match or exceed activation-based vision backbones at lower compute. I am extending this architecture-first program toward new attention and layer primitives and toward recurrence that lets models reason in a learned latent space rather than by stacking depth.

EDUCATION

University of Wisconsin–Madison

Madison, WI

Ph.D., Electrical & Computer Engineering (Focus: Machine Learning)

Aug. 2026 – May 2031

- Advised by Grigorios Chrysos; recipient of the **Distinguished Graduate Fellowship**.

University of Wisconsin–Madison

Madison, WI

B.S. Computer Science (Honors), B.S. Data Science

Aug. 2022 – May 2026

- **GPA: 4.0/4.0**; Dean's List (Fall 2022 – Spring 2025).

PUBLICATIONS

- **Jeffrey Wang**, Jonathan Gregory, Grigorios G. Chrysos. *Activation-Free Backbones for Image Recognition: Polynomial Alternatives within MetaFormer-Style Vision Models*. Proceedings of the 43rd International Conference on Machine Learning (**ICML 2026**). [[code & models](#)]
- Min-Hsuan Yeh, **Jeffrey Wang**, Xuefeng Du, Seongheon Park, Leitian Tao, Shawn Im, Yixuan Li. *Challenges and Future Directions of Data-Centric AI Alignment*. Proceedings of the 42nd International Conference on Machine Learning (**ICML 2025**), Position Paper Track.

RESEARCH EXPERIENCE

University of Wisconsin–Madison

Oct. 2024 – Present

Researcher, PI: Grigorios Chrysos

- First author on **PolyNeXt (ICML 2026)**: a family of activation-free vision backbones replacing ReLU/GELU/softmax with Hadamard-product polynomial modules (PolyMLP, PolyConv, PolyAttn).
- Matched activation-based state-of-the-art (CAFormer) at equal size while staying fully activation-free: **84.3%** top-1 on ImageNet-1K at **26M** params, scaling to **85.2%**.
- Outperformed prior polynomial networks by **2–3%** top-1 at **<half the FLOPs**, with strong out-of-distribution robustness (ImageNet-C/A/R/Sketch) and ADE20K segmentation.
- Ran ImageNet-scale training/eval in PyTorch (DDP, bf16): ablations, robustness, segmentation transfer.
- Built fully-polynomial (BatchNorm) variants using only adds and multiplies, a step toward FHE inference.
- Earlier, designed a DARTS-style NAS method with an entropy penalty that collapses each edge to a single operation, cutting relative error **~15%** vs. standard NAS baselines.

University of Wisconsin–Madison

Jan. 2024 – Sept. 2024

Researcher, PI: Yixuan (Sharon) Li

- Second author on an **ICML 2025** position paper on data-centric AI alignment; designed and ran experiments quantifying how noise in RLHF preference data degrades reward models.
- Designed a noise-robust preference-learning objective (Generalized Binary Cross-Entropy) that improved training stability and accuracy over label-smoothing baselines.

INDUSTRY EXPERIENCE

Amazon Web Services

June 2026 – Aug. 2026

Software Development Engineer Intern (Return)

Seattle, WA

- Building a CloudTrail event de-duplication recommendation system that selects optimal inclusion-key sets to maximize data reduction for IAM Access Analyzer.
- Designed to compress **~100 TB/hr** of event data by **98%+**, cutting downstream storage and analysis cost.
- Engineering the distributed pipeline with Step Functions, Lambda, and a PySpark Glue ETL over Parquet.

Amazon

May 2025 – Aug. 2025

Software Development Engineer Intern

San Diego, CA

- Built a high-throughput DynamoDB CDC to Apache Iceberg/S3 ingestion system (~**100K events/s**) with PySpark/EMR and Kinesis for ML-ready feature stores.
- Delivered a reusable AWS CDK construct to onboard any DynamoDB table with standardized schemas and reconciliation checks, cutting integration time from **weeks to minutes**.
- Hardened reliability and security with CloudFormation/IAM; shipped in TypeScript, Java, and PySpark.

Correkt AI

Aug. 2023 – Aug. 2024

Co-founder & Machine Learning Engineer

Santa Barbara, CA

- Designed and launched a multimodal AI search engine with cited sources and images, serving **40K+ users**; helped secure funding from CNSI, Crescent Fund, and SBAA.
- Fine-tuned LLaMA-3-70B with RLHF; built a synthetic-data pipeline ranking millions of preference pairs at <1% the cost of human annotation, yielding **+10%** preference rate vs. competitors.
- Built web-scale retrieval with Elasticsearch and Groq: **800M+** indexed pages, ~**300ms** avg query latency.

2Sigma School

May 2023 – Aug. 2023

Software Engineer Intern

Remote

- Fine-tuned LLM agents and NLP pipelines for personalized feedback and question generation (cut teacher prep **30%**, **+23%** satisfaction), with code-execution-based hallucination detection.

PROJECTS

Evaluating Modality Bias in Multimodal LLMs

Jan. 2026 – May 2026

Research Project [\[webpage\]](#)

Madison, WI

- Built an ablation pipeline for modality bias in multimodal LLMs (Qwen2.5-Omni, Gemma-3n).
- Found models misreport their modality use and flip **69%** of correct answers under adversarial text injection.

Quantum-Inspired SQL Query Optimization

Sept. 2025 – Dec. 2025

Google-Sponsored Software Engineering Capstone

Madison, WI

- Built a quantum join-reordering pipeline (QAOA, VQE), modeling SQL optimization as a QUBO in Qiskit.
- Achieved **15% lower** average estimated cost vs. the PostgreSQL baseline plan on 5+ table TPC-H queries.
- Implemented SQL-to-QUBO compilation from relational query plans for quantum-annealing optimization.

Depth Efficiency of Recurrent Networks

Jan. 2025 – May 2025

Research Project

Madison, WI

- Extended depth-efficiency theory (why deeper nets are exponentially more expressive) to recurrent models.
- Proved a Tensor-Train network is RNN-equivalent with the same exponential rank separation, via measure theory.

HONORS & AWARDS

- Distinguished Graduate Fellowship, University of Wisconsin–Madison (2026)
- Dean’s List (Fall 2022 – Spring 2025), University of Wisconsin–Madison
- USACO Gold (2021)

SERVICE

- Reviewer, ICLR 2026 Workshop on Agentic AI in the Wild: From Hallucinations to Reliable Autonomy

TECHNICAL SKILLS

- **Machine Learning:** Deep Learning, Transformers, Computer Vision, NLP, RLHF, Fine-tuning, Neural Architecture Search, RAG, Distributed Training
- **Frameworks:** PyTorch, Hugging Face, DeepSpeed, vLLM, TensorFlow, NumPy, Pandas, CUDA, Qiskit
- **Cloud / Data:** AWS (Lambda, Glue, EMR, Kinesis, DynamoDB, CDK), GCP, Spark, Apache Iceberg
- **Backend:** Django, Flask, Elasticsearch, MongoDB, REST APIs
- **Languages:** Python, C/C++, Java, TypeScript, JavaScript, SQL
- **Tools:** Docker, Git, Linux, Slurm, HTCondor, CI/CD, CloudFormation, Weights & Biases